MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/78-RG217

THIS CABLES USES PVC MATERIAL AND IS NOT TO BE USED IN AEROSPACE APPLICATIONS.

NOTE: THE AIR FORCE HAS RESTRICTED THE USE OF PVC IN AEROSPACE AND GROUND SUPPORT APPLICATIONS. CABLES WITH PVC JACKETING SHALL BE USED FOR RETROFIT PURPOSES ONLY UNTIL AN ALTERNATE JACKET IS APPROVED.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-17.

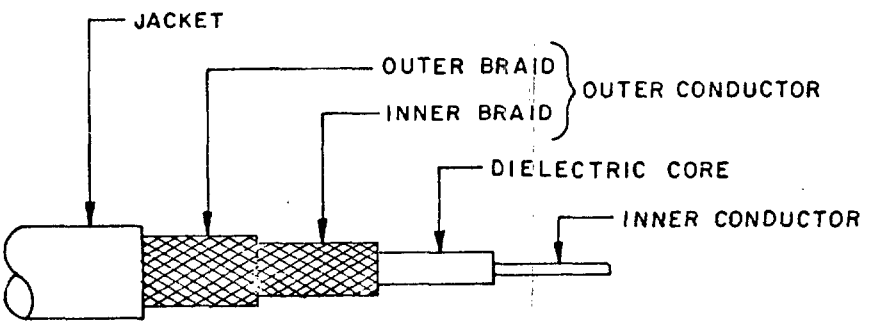


FIGURE 1. Configuration.

Ⓑ denotes changes

FSC 6145

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TABLE I. Description.

Component	Construction details
Inner conductor	Solid bare copper wire. Diameter: 0.106 inch ±0.001.
Dielectric core	Type A-1: Solid polyethylene Diameter: 0.370 inch ±0.010.
Outer conductor	Double braid of AWG No. 33 bare copper wire. Diameter: 0.463 inch maximum.
Inner braid	Coverage : 94.8% nominal Carriers : 24 Ends : 10 Picks/inch: 5.4 ±10%
Outer braid	Coverage : 93.6% nominal Carriers : 24 Ends : 8 Picks/inch: 10.6 ±10%
Jacket	Type IIa: PVC. Diameter: 0.545 inch ±0.010.

ENGINEERING INFORMATION:

Continuous working voltage: 5,800 V rms, maximum.

Operating frequency: 3 GHz maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -40°C to +85°C.

Inner conductor properties:

DC resistance (maximum at 20°C): 0.0941 ohm per 100 feet.

(B) Elongation: 30 percent, minimum.

Tensile strength: Not applicable.

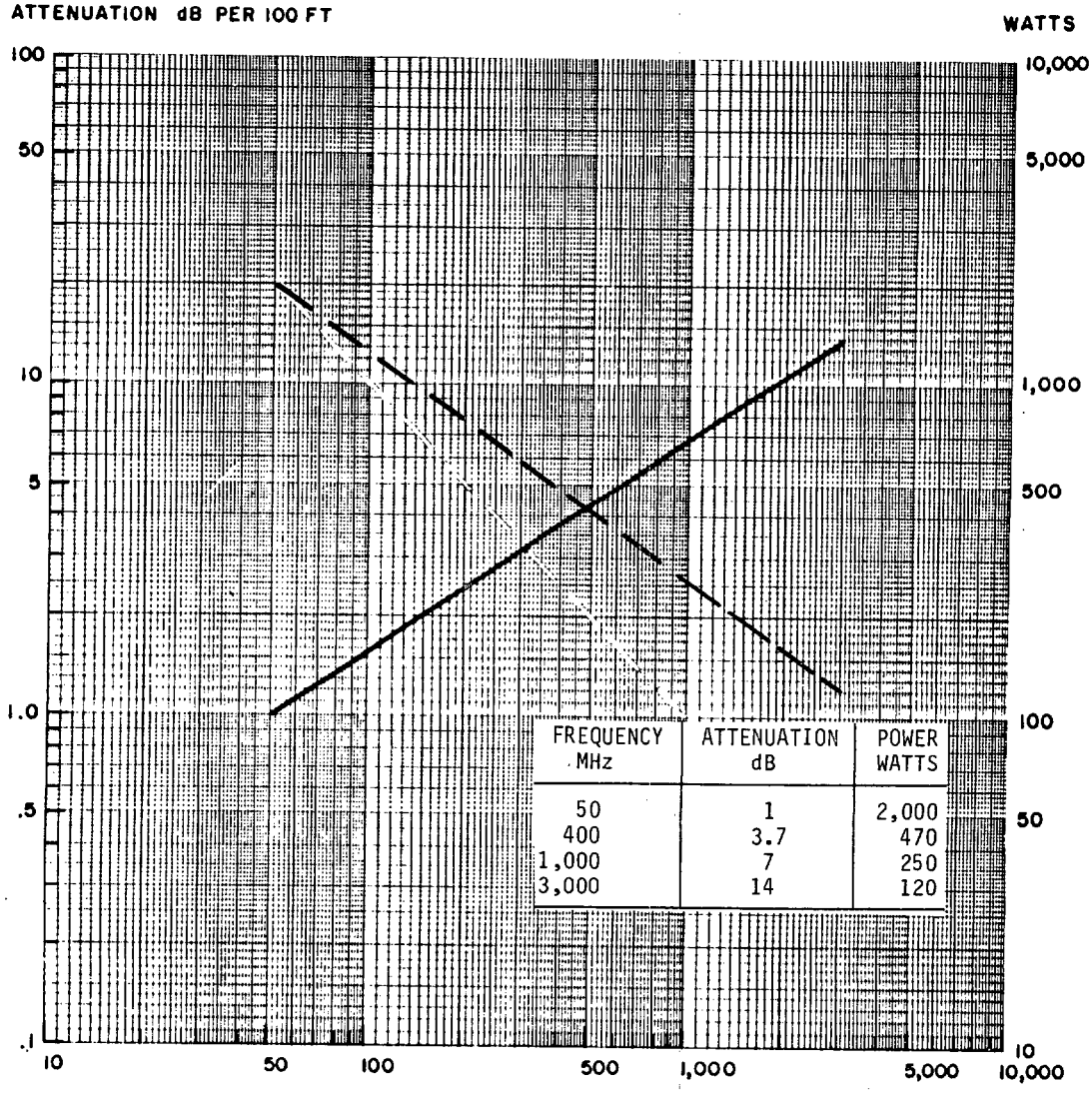
Engineering note: This cable is useful in general purpose, medium low temperature applications. (See connector series "N" and "SC" per MIL-C-39012.)

REQUIREMENTS:

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

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FREQUENCY MHz	ATTENUATION dB	POWER WATTS
50	1	2,000
400	3.7	470
1,000	7	250
3,000	14	120

MAXIMUM ATTENUATION ———

MAXIMUM POWER — — —

AT 25°C SEA LEVEL

Tabulated values are for reference only.
The values on the chart represent the requirements.

FIGURE 2. Power rating and attenuation.

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Visual and mechanical examination:

Out-of-roundness: Applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors:

(B) Inner conductor to core: 15 pounds, minimum; 150 pounds, maximum.

Aging stability: +98°C ±2°C.

Stress crack resistance: Not applicable.

Outer conductor integrity: Not applicable.

(B) Cold bend: -40°C ±2°C.

Dimensional stability: +85°C ±2°C.

Inner conductor from core: 0.125 inch, maximum.

Inner conductor from jacket: 0.250 inch, maximum.

Contamination: Applicable.

Bendability: Not applicable.

Flammability: Not applicable.

(B) Weight: 0.225 pound per foot, maximum.

Electrical:

Continuity: Applicable.

(B) Spark test: 8,000 V rms, +10%, -0%.

(B) Voltage withstanding: 12,000 V rms, +10%, -0%.

Insulation resistance: Not applicable.

Corona extinction voltage: 7,000 V rms, minimum.

Characteristic impedance: 50 ±2 ohms.

Attenuation: See figure 2.

Structural return loss: See figure 3.

(B) Capacitance: 32.2 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Part number: M17/78-RG217.

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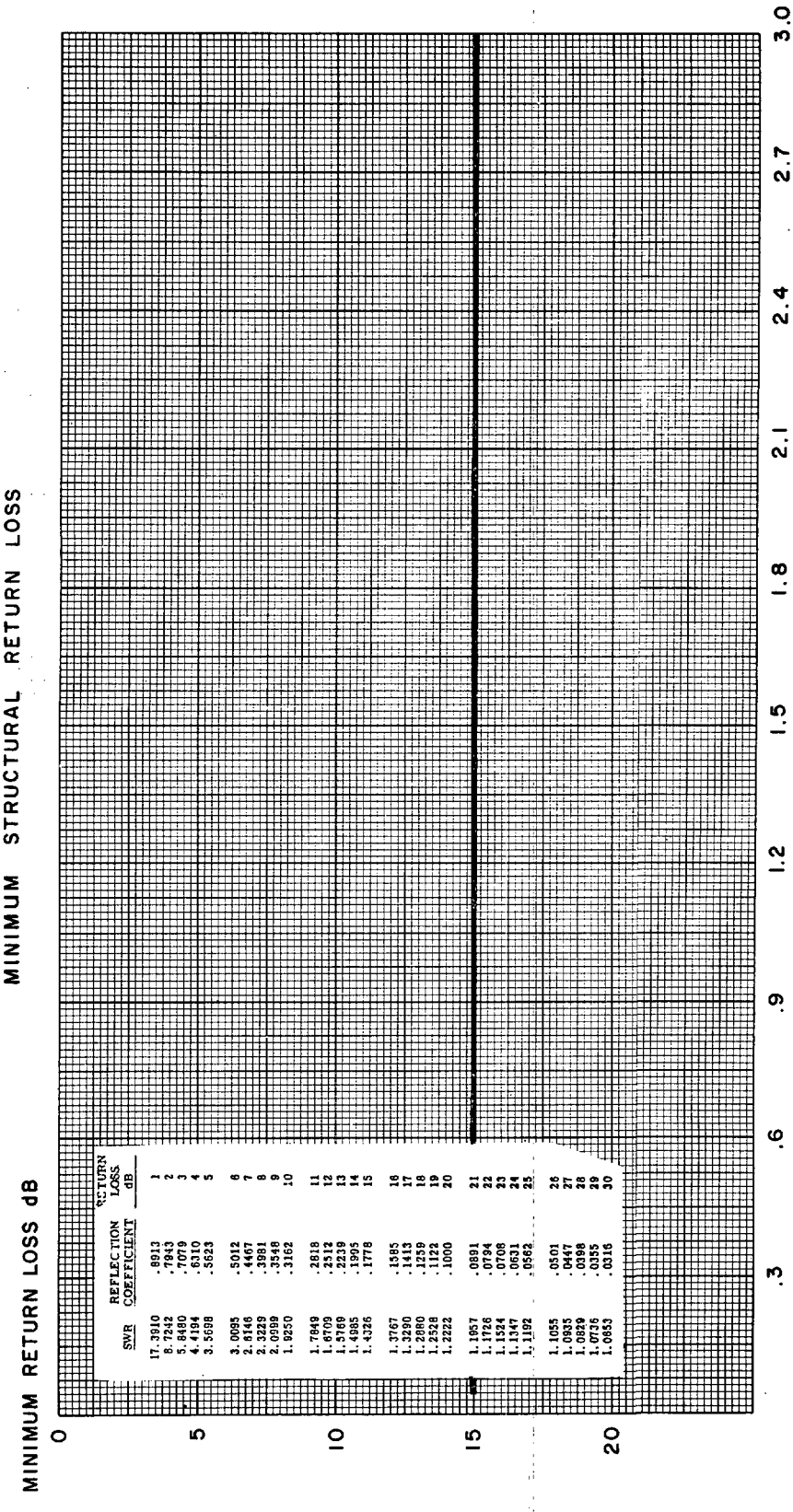


FIGURE 3. Structural return loss.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, TD
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - AR, AT, ME
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

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